MIL-P-17303(SHIPS)
2 September 1952

INTERIM MILITARY SPECIFICATION

PACKING MATERIAL, METALLIC AND NONMETALLIC

1. SCOPE

1.1 Scope. - This specification covers plastic metallic packings for rotary and reciprocating pumps handling steam, water, gas and oil; and plastic nonmetallic packings for high and low pressure valve stem and reciprocating rod saturated and superheated steam service for machinery on Naval vessels.

1.2 Classification. - Plastic packing shall be of the following classes and types, as specified (see 6.1):

Class I - Plastic metallic, lead base:
Type A - Bulk form, symbol 1431.
Type B - Coil form, with or without cotton jacket, symbol 1433.
Type C - Coil form, metal foil jacket, symbol 1434.
Type D - Coil form, braided metal jacket, symbol 1432.
Class II - Plastic nonmetallic:
Type B - Coil form, cotton jacket, symbol 1106.
Type C - Coil form, asbestos and wire jacket, symbol 1108.
Class III - Plastic metallic, copper base:
Type B - Coil form, cotton jacket, symbol 1439.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
NN-B-591 - Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601 - Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
NN-B-631 - Boxes, Wood, Wirebound (for Domestic Shipment).
LLL-B-631 - Boxes, Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment).

MILITARY
JAN-P-105 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed.
MIL-B-107 - Boxes, Wood, Wirebound (Overseas Type).
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN-P-125 - Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible.
JAN-P-138 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Fiberboard-Lined.
MIL-A-140 - Adhesive, Water-Resistant, Waterproof Barrier-Material.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

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STANDARDS

MILITARY

MIL-STD-129 - Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. - Packing material furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4 (see 6.2).

3.2 Composition and construction. - The composition and construction shall conform to that of the sample submitted for qualification (see 4.4.1.2).

3.3 Size. - Packing shall be furnished in the sizes specified (see 6.1).

3.4 Binder. - The different components of type A, and of the core of types B, C, and D shall be thoroughly mixed with a binder suitable for the temperature conditions for which the packing is required; and shall be so held together by the binder as to prevent separation of the various parts.

3.5 Graphite. - Graphite, when used for lubrication, shall contain not less than 90 percent, by weight, of graphitic carbon (see 4.5.2).

3.6 Flexibility. - The packing shall be sufficiently flexible to be readily formed into coils to fit a rod of the diameter shown in table I (see 4.5.3).

Table I - Size of coils.

Size of packing	Rod diameter
Inches	Inches
1/8 to 7/16, inclusive	1
1/2 and 5/8	2
11/16 to 13/16, inclusive	4
7/8 and 1	8
1- 1/8	9
1- 1/4	10

3.7 Simulated performance. - The packaging under initial conditions and during the progress of the simulated service tests specified in 4.5.4 shall respond readily to gland adjustment for the control of leakages from the stuffing box. The packing shall be in satisfactory condition upon completion of the tests.

3.8 Class I, plastic metallic, lead base. - The metal used for lead base packing shall not melt nor disintegrate at or below a temperature of 550° F. (see 4.5.5).

3.8.1 Type A, symbol 1431. - The packing shall be in bulk form.

3.8.2 Type B, symbol 1433. - The packing shall be continuous strips of approximately uniform cross section, partially compressed to prevent disintegration during application. It may be enclosed in a latticed cotton thread or knit cotton covering or may be furnished without a covering.

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3.8.3 Type C, symbol 1434. - The packing shall be continuous strips of uniform cross section, completely enclosed in a metal covering consisting of not less than two ribbons. The metal ribbon shall be not less than 0.0023 inch thick.

3.8.4 Type D, symbol 1432. - The packing shall be continuous strips of uniform cross section, extruded into a braided wire jacket. The jacket shall be of copper wire not more than 0.008 inch in diameter.

3.9 Class II, plastic nonmetallic. -

3.9.1 Type B, symbol 1106. - The packing shall be continuous strips of approximately uniform cross section, partially compressed to prevent disintegration during application, and enclosed in a latticed cotton thread or knit cotton covering.

3.9.2 Type C, symbol 1108. - The packing shall be continuous strips of uniform cross section, completely enclosed in a braided jacket composed of asbestos and nickel-copper wire.

3.9.2.1 Corrosion resistance. - Symbol 1108 packing shall not corrode 13 percent chrome steel valve stem when tested as specified in 4.5.6.

3.10 Class III, plastic metallic, copper base. - The metal for class III packing shall not melt nor disintegrate at or below a temperature of 1200° F. (see 4.5.5).

3.10.1 Type B, symbol 1439. - The packing shall be in continuous strips of approximately uniform cross section, partially compressed to prevent disintegration during application, and enclosed in a latticed cotton thread or knit cotton covering.

3.11 Workmanship. - The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Qualification tests at a Government laboratory. - Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.5.

4.3 Sampling. -

4.3.1 Lot. - For purposes of sampling, a lot shall consist of all packing of one class and type produced in one plant under essentially the same conditions and offered for delivery at one time.

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4.3.2 Sampling procedure for lot acceptance inspection. - For the inspection specified in 4.4.1.1, the Government inspector shall select representative lengths of packing from each lot in accordance with table II. When the order calls for more than one size, samples shall be selected from the smallest and from the largest sizes.

Table II - Sampling procedure for lot acceptance inspection.

Number of lengths of packing in the lot	Number of sample lengths to be selected
8 or under	2
9 to 15	3
16 to 25	5
26 to 40	7
Over 40	10

4.3.3 Sampling procedure for lot acceptance tests. - For the test specified in 4.4.1.2 the Government inspector shall take a 36-inch sample from each sample length of finished packing selected in accordance with table II. When the order calls for more than one size, samples shall be selected from the smallest and from the largest sizes.

4.3.4 Sampling procedure for production check tests. - From the first lot of packing of each class and type offered for delivery under a contract or order, the Government inspector shall select from each of two lengths of packing a 36-inch sample for the purpose of the production check tests specified in 4.4.2. Thenceforth, two such sample pieces of packing shall be selected from one of every 10 lots which have passed inspection and test at the place of manufacture.

4.4 Inspection and tests. -

4.4.1 Lot acceptance inspection and tests (at place of manufacture). -

4.4.1.1 Inspection. - Each of the samples selected in accordance with 4.3.2 shall be subjected by the Government inspector to surface inspection for workmanship and dimensions. Minor surface defects not affecting the serviceability of the packing shall not be cause for rejection.

4.4.1.2 Tests. - The samples selected in accordance with 4.3.3 shall be subjected to those tests which will establish the identity of the inspection sample with that given qualification and to the tests specified in 4.5.3 and 4.5.5.

4.4.1.3 Action in case of failure. - If any one of the samples tested is found to be not in conformance with the specification, the lot which it represents shall be rejected. A rejected lot may be resubmitted for Government inspection only after the manufacturer, after being informed of the reasons for rejectance, has so reworked the entire lot as to remove or correct all nonconforming material.

4.4.2 Production check tests (at a Government laboratory). - The samples selected in accordance with 4.3.4 shall be subjected to the tests specified in 4.5.2, 4.5.3, and 4.5.5. If found not to conform as to construction with that given qualification, the sample need not be subjected to other tests before reporting the lot as unsatisfactory.

4.4.2.1 Place of tests. - Unless otherwise specified, production check tests shall be conducted at the Naval Engineering Experiment Station, Annapolis, Md.

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4.4.2.2 Action in case of failure. - Acceptance of the first lot offered for delivery under a contract or order shall be withheld until a satisfactory report is received on the production check test sample. Thenceforth, except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the sampling and inspection specified in 4.3.2, 4.3.3 and 4.4.1 and acceptance shall not be withheld pending receipt of test reports on production check test samples. However, upon receipt of an unsatisfactory test report on a production check test sample, the Government inspector shall select additional samples from every subsequent lot offered for delivery. The samples so selected shall be submitted to Naval Engineering Experiment Station, Annapolis, Md., and shall there be subjected to test or tests wherein failure was observed. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. Additional testing shall be discontinued and lot acceptance returned to the normal basis when three successive lots have been accepted.

4.5 Test procedures. -

4.5.1 Chemical analysis. - Chemical analysis shall be made using standardized procedures.

4.5.2 Graphite carbon, by combustion method. - Place a sample weighing 2 to 3 grams in a 250-milliliter (mℓ) Erlenmeyer flask fitted with a reflux condenser, add 10 grams of para-nitrotoluene and 50 mℓs of ortho-dichlorobenzene and allow to reflux for 24 hours or until all the binder is in solution. Allow to cool and add 100 mℓs of chloroform, filter on No. 42 filter paper, wash with chloroform several times and dry in the oven in a current of air at 105° Centigrade (C.). Treat the ignited residue with concentrated hydrochloric acid overnight or until all metal is in solution, filter and wash with distilled water until free from chlorides. Weigh 0.200 to 0.300 gram of the pure graphite in a porcelain boat and place in the tube of the combustion furnace and burn at 800-850° C. in a stream of oxygen gas. The CO_2 which forms is collected in an ascarite filled absorption tube. The combustion tube shall contain some fused lead chromate to retain any sulfur that may be present.

$$\text{Calculations: } \frac{\text{Weight of } CO_2 \times \text{Factor } (0.2729)}{\text{Weight of sample}} \times 100 = \%C.$$

4.5.3 Flexibility. - Packing shall be coiled by hand around a rod of a diameter as shown in table I (see 3.6). Only sufficient packing to make one complete turn around the rod shall be used.

4.5.4 Simulated performance. - Performance tests shall be made under simulated operating conditions in a machine designed for the purpose.

4.5.4.1 Type C, symbol 1108, and type B, symbol 1439. - The packing shall be tested for the following service:

Valve stem service at 600 pounds per square inch (p.s.i.) and 750° F. Test shall be concluded after 3500 hours of operation.

4.5.4.2 Type D, symbol 1432. - The packing shall be tested for the following service:

Reciprocating rod service with steam at 300 p.s.i. and 720° F., and a rod speed of 170 feet per minute. The test shall be concluded after 1500 hours operation.

4.5.4.3 Type A, symbol 1431, type B, symbol 1106 and type B, symbol 1433. - The packing shall be tested for the following services:

Reciprocating rod service with saturated steam at 300 p.s.i. and a rod speed of 100 feet per minute. Test shall be included after 3000 hours of operation.
Rotary rod service with feed water at 100 p.s.i. gage and 250° F., and a rod speed of 4000 feet per minute. The test shall be concluded after 3000 hours of operation.

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4.5.4.4 Type C, symbol 1434. - The packing shall be tested for the following service:

Rotary rod service with feed water at 100 pounds p.s.i. gage and 250° F., and a rod speed of 4000 feet per minute. Test shall be concluded after 3000 hours of operation.

4.5.4.5 Rotary rod service. - During the rotary rod service tests specified in 4.5.4.3 and 4.5.4.4 in order to simulate conditions occasionally encountered in the service, the packing shall be subjected to cycles of sudden changes in pressure from 100 p.s.i. to 10 p.s.i. twice daily. The lower pressure shall be maintained for approximately 30 seconds and then suddenly raised to 100 p.s.i.

4.5.5 Heat resistance. - Specimens cut from each sample of packing shall be subjected for 2 hours to the temperature specified in 3.8 and 3.10.

4.5.6 Corrosion resistance, symbol 1108, packing. - The packing shall be installed in the stuffing box of 1-inch size, 600 p.s.i. valve, equipped with a 13 percent chrome steel valve stem. The packing gland shall be adjusted to allow slight leakage under 300 p.s.i. boiler feed water. The condition shall be maintained for 24 hours to insure the wetting of the packing. The valve shall then be drained and stored at ambient temperature. The valve stem shall be inspected after 6 and 12 months of storage.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Commercial packages are acceptable under this specification.

5.2 Packing. -

5.2.1 For domestic shipment. - The subject commodity packaged as specified in 5.1, shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery. When wood-cleated-fiberboard boxes, wood-cleated-plywood boxes, nailed wood boxes, wirebound wood boxes, corrugated fiber boxes, or solid fiber boxes are used, they shall conform to Specifications NN-B-591, NN-B-601, NN-B-621, NN-B-631, LLL-B-631 and LLL-B-636, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

5.2.2 For overseas shipment. - The subject commodity packaged as specified in 5.1, shall be packed in cleated-plywood boxes, nailed wood boxes, wirebound boxes, fiberboard boxes, or fiberlined wood boxes conforming to Specifications JAN-P-105, JAN-P-106, MIL-B-107, JAN-P-108, and JAN-P-138, respectively. Shipping containers shall be lined with a sealed waterproof bag made from material conforming to Specification JAN-P-125. The seams and closures shall be sealed with water-resistant adhesive conforming to Specification MIL-A-140. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

5.3 Marking. - In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, type and size required (see 1.2 and 3.3).
(c) Whether domestic or overseas shipment is required (see 5.2.1 and 5.2.2).

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6.2 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of Ships, Navy Department, Washington 25, D. C., and arrange to have the products that they propose to offer to the Army, the Navy, or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Navy Department, Washington 25, D. C.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in anv way be related thereto.

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